TEXTBOOK EXERCISES

I. Choose the correct answer.

1. Which of the following vibrates when a musical note is produced by the cymbals in a orchestra?
   a) stretched strings
   b) stretched membranes
   c) air columns
   d) metal plates
2. Sound travels in air:
   a) if there is no moisture in the atmosphere.
   b) if particles of medium travel from one place to another.
   c) if both particles as well as disturbance move from one place to another.
   d) if disturbance moves.
3. A musical instrument is producing continuous note. This note cannot be heard by a person having a normal hearing range. This note must then be passing through
   a) wax b) vacuum
   c) water d) empty vessel
4. The maximum speed of vibrations which produces audible sound will be in
   a) sea water b) ground glass
   b) dry air d) human blood
5. The sound waves travel faster
   a) in liquids b) in gases
   c) in solids d) in vacuum

II. Fill in the blanks.

1. Sound is a _____ wave and needs a material medium to travel.
2. Number of vibrations produced in one second is _____ .
3. The velocity of sound in solid is _____ than the velocity of sound in air.
4. Vibration of object produces _____ .
5. Loudness is proportional to the square of the _____ .
6. _____ is a medical instrument used for listening to sounds produced in the body.
7. The repeated reflection that results in persistence of sound is called _____ .

III. Match the following.

Tuning fork	The point where density of air is maximum.
Sound	Maximum displacement from the equilibrium position.
Compressions	The sound whose frequency is greater than 20,000 Hz.
Amplitude	Longitudinal wave.
Ultasonics	Production of sound.

IV. Answer briefly.

1. Through which medium sound travels faster, iron or water? Give reason.
2. Name the physical quantity whose SI unit is 'hertz'. Define.
3. What is meant by supersonic speed?
4. How does the sound produced by a vibrating object in a medium reach your ears?
5. You and your friend are on the moon. Will you be able to hear any sound produced by your friend?

V. Answer in detail.

1. Describe with diagram, how compressions and rarefactions are produced.
2. Verify experimentally the laws reflection of sound.
3. List the applications of sound.
4. Explain how does SONAR work?

IX_SCI_EM_Unit-08_PHY.indd 100 12/7/2022 4:21:50 PM